

6.19 Solution: (a) Space inversion can be achieved in spherical coordinates by setting $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow -\phi$. Then, the only non-zero component of the vector potential becomes

$$\mathbf{A} = \frac{g}{4\pi r} \frac{1 - \cos \theta}{\sin \theta} \hat{\phi} \rightarrow \mathbf{A}' = \frac{g}{4\pi r} \frac{1 - \cos(\pi - \theta)}{\sin(\pi - \theta)} (-\hat{\phi}) = -\frac{g}{4\pi r} \frac{1 + \cos \theta}{\sin \theta} \hat{\phi}$$

By taking the curl of the vector potential, it can be shown that this leads to the same magnetic field as in Problem 6.18 (b), except that now the magnetic field is not defined at $\theta = 0$.

(b) It is straightforward to get the difference as

$$\delta \mathbf{A} = \mathbf{A}' - \mathbf{A} = -\frac{g}{2\pi r \sin \theta} \hat{\phi}.$$

In spherical coordinates, the gradient operator is

$$\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi},$$

we can see that, by setting

$$\chi = -\frac{g}{2\pi} \phi,$$

we will have $\delta \mathbf{A} = \nabla \chi$.

(c) For the two formulations of the vector potential, we have

$$\mathbf{A}(\mathbf{x}) = \frac{g}{4\pi} \int_L \frac{d\mathbf{l}' \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3},$$

where L is from $-\infty$ to the origin along the negative z -axis in Problem 6.18 and from $+\infty$ to the origin along the z -axis in this problem. Then,

$$\delta \mathbf{A} = \mathbf{A} - \mathbf{A}' = \frac{g}{4\pi} \int_{-\infty}^{+\infty} \frac{\hat{k} \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} dz' = \frac{g}{2\pi} \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2} = \frac{g}{2\pi r \sin \theta} \hat{\phi} = \nabla \chi,$$

with

$$\chi = \frac{g}{2\pi} \phi.$$

Therefore, the gauge function comes from the choice of integration contour.